

Jung Ryul Lee Multi-Asset Portfolio

9.5% Expected return (1y, ex-ante, KRW)	5.3% Volatility (benchmark* 8.7%)	1.26 Sharpe vs 2.8% KRW cash (benchmark* 0.41)	0.21 / 0.11 Beta to S&P 500 / KOSPI 200 (in KRW)	300 → 19 Stocks screened → stocks held
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Holdings, weights and rationale

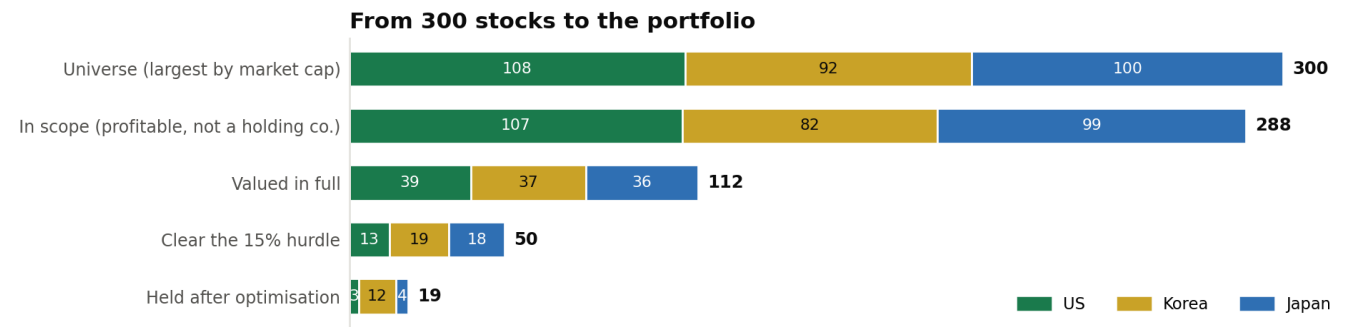
Holding	Sleeve	Weight	Why it is held	Valuation / return source
United States · 3 stocks, 13 ETFs				
IVV iShares Core S&P 500 ETF	Growth	5%	Core US beta; mega-caps at index weight	Earnings yld + growth
USMV iShares MSCI USA Min Vol Factor ETF	Growth	1%	Defensive US equity (low-vol factor)	Low-vol premium
XLU Utilities Select Sector SPDR ETF	Growth	1%	Defensive equity; AI power demand	Defensive yield
T AT&T	Growth	2%	10% FCF yield, low beta	FCFE +40%
CI Cigna Group	Growth	1.75%	PBM + insurer at 8.6x forward P/E	FCFE +55%
VNQ Vanguard Real Estate ETF	Real asset	1%	Broad US REITs	Earnings yld + growth
VICI VICI Properties	Real asset	1.25%	Experiential net lease, 9.4x AFFO	AFFO model +41%
IEF iShares 7-10Y Treasury ETF	Hedge	2.5%	Recession / deflation hedge	YTM ≈ 5.0%
VTIP Vanguard Short-Term TIPS ETF	Hedge	2.75%	Inflation hedge, short duration	Real yld + CPI
SGOV iShares 0-3M T-Bill ETF	Hedge	3.25%	USD cash; dry powder	≈4.2%
IAU iShares Gold Trust	Hedge	1.5%	Geopolitics / Fed-credibility hedge	Diversifier
DBMF iMGP DBi Managed Futures ETF	Hedge	3.75%	Trend-following; stocks & bonds down	Trend premium
KMLM KFA Mount Lucas Managed Futures ETF	Hedge	3.25%	Second trend model (diversifies DBMF)	Trend premium
TAIL Cambria Tail Risk ETF	Hedge	4.5%	Crash insurance (puts + T-notes)	Convexity
BTAL AGF US Market Neutral Anti-Beta ETF	Hedge	4.5%	Anti-beta: gains when high-beta falls	Negative beta
PDBC Invesco Optimum Yield Diversified Commodity ETF	Hedge	1.5%	Oil / commodity-shock hedge	Roll + collateral
Korea · 12 stocks, 8 ETFs				
069500 KODEX 200	Growth	2%	Core Korea beta	Earnings yld + growth
316140 Woori Financial Group	Growth	2%	Bank at 0.59x book, 3.9% dividend	Justified P/B +69%
015760 Korea Electric Power	Growth	2.75%	Utility at 0.40x book	Justified P/B +118%
033780 KT&G	Growth	2.75%	Defensive staples, 15% ROE	Justified P/B +28%
024110 Industrial Bank of Korea	Growth	2.25%	Policy bank at 0.46x book, 4.9% dividend	Justified P/B +101%
086280 Hyundai Glovis	Growth	1.5%	Logistics, 15% ROE	Justified P/B +43%
030200 KT Corp	Growth	2.5%	Telco at 0.63x book, 4.5% dividend	Earnings/FCF model +61%
005830 DB Insurance	Growth	2.75%	Non-life insurer, 16% ROE at 1.0x book	Justified P/B +101%
259960 Krafon	Growth	2.5%	Game developer at 8.5x fwd P/E, net cash	Earnings/FCF model +98%
021240 Coway	Growth	2.5%	Appliance rental, 19% ROE at 9x P/E	Earnings model +80%
161390 Hankook Tire & Technology	Growth	2.5%	Tyres at 5.4x fwd P/E, 0.49x book	Earnings/FCF model +61%
032640 LG Uplus	Growth	2.5%	Telco at 0.69x book, 4.5% dividend	Earnings/FCF model +79%
175330 JB Financial Group	Growth	2%	Regional bank, 12% ROE at 0.86x book	Justified P/B +82%
329200 TIGER REITs & Real Estate Infra	Real asset	2.25%	K-REITs & infra, ~9.7% distribution	Distribution ≈9.7%
152380 KODEX 10Y KTB Futures	Hedge	5.75%	KRW duration hedge	YTM ≈ 4.4%
114260 KODEX 3Y KTB	Hedge	1.5%	Short KRW duration	YTM ≈ 3.3%
261240 KODEX USD Futures	Hedge	1%	Extra USD vs KRW hedge	FX (carry -1.2%)
459580 KODEX CD Rate Active	Hedge	1%	KRW cash; liquidity	CD rate
132030 KODEX Gold Futures (H)	Hedge	1.5%	Gold, FX-hedged	Diversifier
114800 KODEX Inverse (KOSPI 200)	Hedge	1%	Direct Korea crash hedge	Negative beta
Japan · 4 stocks, 3 ETFs				
1306 NEXT FUNDS TOPIX ETF	Growth	2%	Core Japan beta + yen	Earnings yld + growth
8725 MS&AD Insurance	Growth	2.25%	Non-life insurer, 16% ROE at 1.06x book	Justified P/B +83%
8591 ORIX	Growth	2%	Diversified finance at 10.8x P/E	Justified P/B +55%
8630 Sompo Holdings	Growth	2.25%	Non-life insurer, 14% ROE at 1.1x book	Justified P/B +71%
4503 Astellas Pharma	Growth	2.25%	Pharma at 11.6x fwd P/E, 21% ROE	Earnings/FCF model +54%
1343 NEXT FUNDS REIT Index ETF	Real asset	1.5%	J-REITs, rent reflation	Distribution ≈4.5% (est.)
2561 iShares Core Japan Govt Bond ETF	Hedge	2.75%	Yen safe haven + JGB carry	Yen + ≈2.5% yield
Total		100.0%		

*Benchmark: 30% IVV / 30% KODEX 200 / 20% IEF / 20% KODEX KTB10, in KRW. Expected returns are model outputs, not forecasts. Toyota is no longer held (page 4). Korean and Japanese stocks are listed by their exchange codes on page 5.

1 · Thesis, regime and how the stocks were chosen

Thesis. (1) With the US 10-year at 5.23% and the JGB 10-year at 3.07%, discount rates are high, so single-stock risk goes only where a conservative valuation still shows ≥15% upside. (2) The earlier version valued a hand-picked shortlist of 50 names, which invites selection bias. This version starts from the **300 largest stocks** in the US, Korea and Japan, screens them on multiples, values 112 in full and lets a resampled optimiser choose among the survivors. (3) Hedges are split by type of crisis and by currency: for a KRW investor, the dollar and the yen have historically risen when Korean assets fall.

Indicator (late Sep 2026)	Level	Implication
US 10-yr / 2-yr / Fed funds	5.23% / 4.91% / 4.00%	High discount rate. Long-duration equity must clear a ~9.5–10% cost of equity.
Korea 10-yr KTB	4.41% (+1.56pp y/y)	Korean cost of equity 8.8–13% depending on beta; banks, telcos and insurers still clear it.
S&P PMI / WTI / Gold	58.4 / \$94 / \$4,266	Overheating + oil risk: the nearer tail is inflation, so hedges must work when bonds fail.
JGB 10-yr / BOJ policy rate	3.07% / 1.25%	Highest JGB yield since 1996. Japanese insurers and lenders earn more on their float.
Samsung / SK Hynix	₩286,500 / ₩1.86m	Korea's index is riding an AI-memory boom; the screen deliberately values memory on mid-cycle returns.

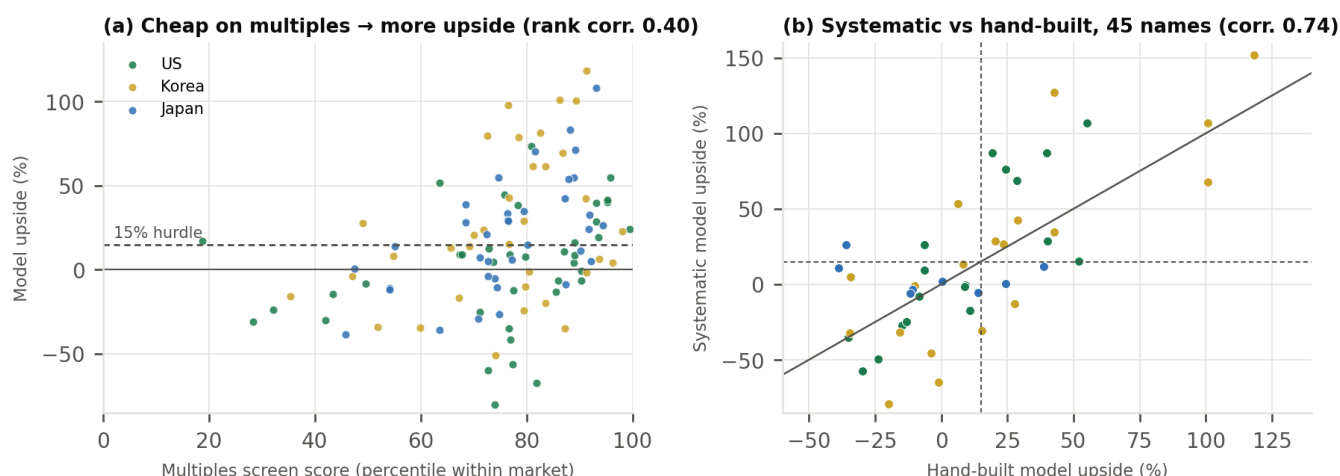


1. Universe	The ~100 largest listed companies by market cap in each market (108 US, 92 Korea [KOSPI; KOSDAQ names were not in the data feed], 100 Japan). Multiples from Stock Analysis, 22–28 Sep 2026.
2. Scope	Out: loss-makers, NAV-type holding companies (SK Square, SK Inc., SoftBank Group, LG Corp, Samsung C&T ...) and one REIT with no AFFO data. 288 remain.
3. Screen	Within each market, rank on three yields: forward earnings yield, free-cash-flow yield and ROE ÷ P/B. The top 30 per market, plus every name that already had a hand-built model, are valued in full (112 names).
4. Value	One model per business type (page 3). Hand-built models from the earlier version are kept where they exist.
5. Hurdle	Held only if the base case shows ≥15% upside: 50 pass (13 US, 19 Korea, 18 Japan).
6. Select	60 max-Sharpe runs with noisy expected returns; a stock is kept if it gets ≥0.5% in at least half of them (page 4). 19 survive. No market quotas: the split is whatever the screen produces.

2 · The valuation models and whether they can be trusted

Every model uses a CAPM cost of equity: US 5.23% + $\beta \times 4.5\%$, Korea 4.41% + $\beta \times 5.5\%$, Japan 3.07% + $\beta \times 5.0\%$, with β floored at 0.8 and capped at 1.6 so that no stock gets a cost of equity below its market's. Terminal growth is 3% (US), 2.5% (Korea) and 1.5% (Japan).

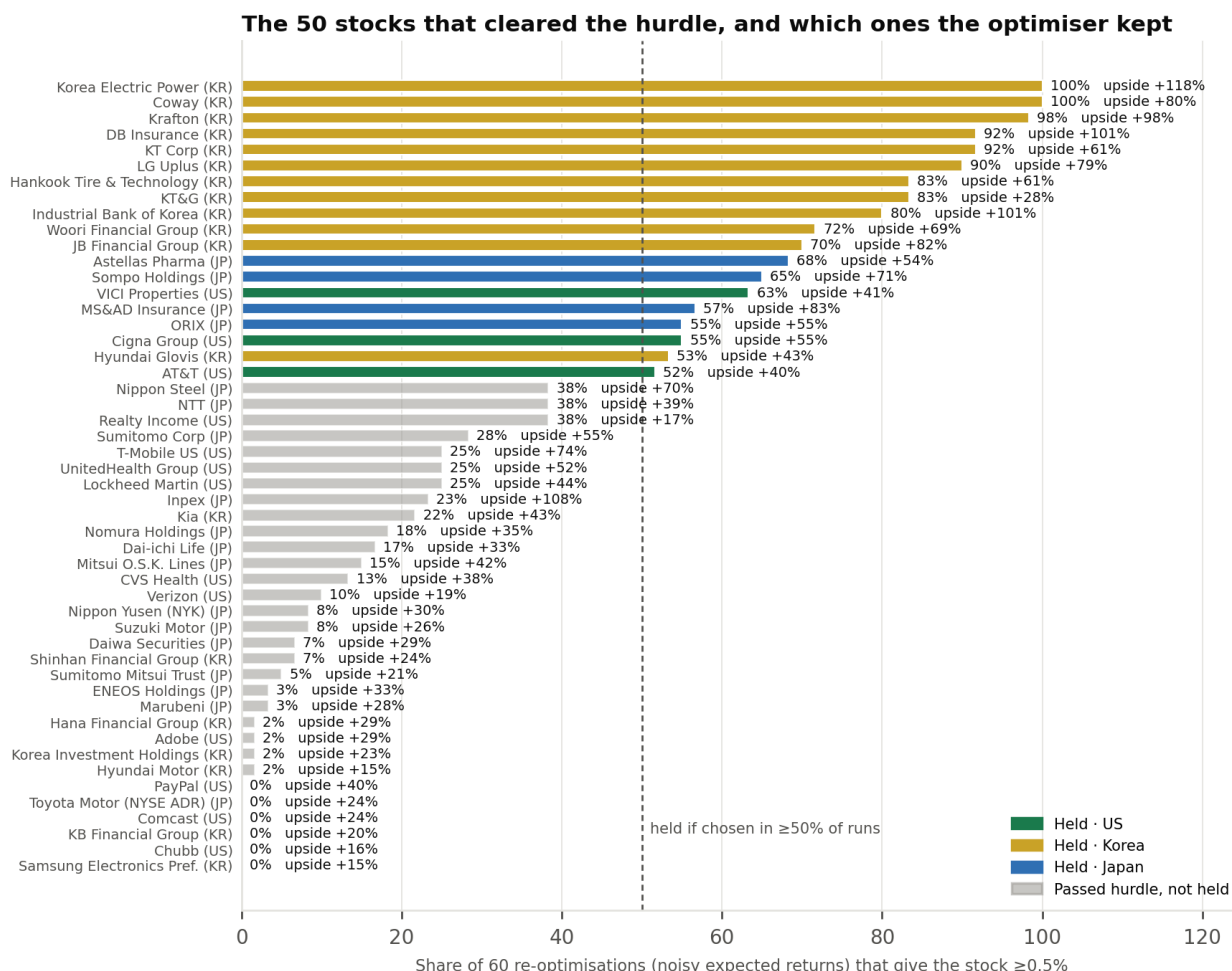
Business type	Model	Key assumptions	Names valued
Banks, insurers, brokers	Justified P/B = $(ROE - g)/(k - g)$	ROE = average of trailing and forward-implied (P/B + fwd P/E). Brokers capped at 12% ROE because 2026 trading profits are a boom.	34
Cyclicals	Justified P/B on a mid-cycle ROE	Memory 14%, oil & gas 11%, shipbuilding 11%, refining 8%, container shipping 8%, steel/metals and commodity chemicals 7%, airline 7%. Peak earnings are ignored.	15
Everything else	10-yr earnings model, blended 50/50 with a free-cash-flow model	Year-1 = forward EPS. Growth = half the consensus 3-yr EPS growth + half terminal, capped at 10% and at $0.6 \times ROE$; 5 years, then a 5-year fade. Payout = $1 - g/ROE$. The FCF leg is never credited above earnings. Shrinking firms get no terminal growth.	60
Hand-built (earlier version)	FCFF/FCFE DCF, cycle DCF, AFFO, justified P/B or P/E	Name-specific inputs (normalised margins, ROE, AFFO). Used instead of the systematic value wherever it exists.	49



Three checks. (a) Stocks that screen cheap on multiples do show more model upside, but the rank correlation is only moderate, which is the point of valuing them properly rather than buying the cheapest decile. (b) On the 45 names that have both a hand-built and a systematic value, the two agree reasonably (correlation 0.74, median gap +2pp) and land on the same side of the 15% hurdle 84% of the time. The biggest disagreements are KT&G (hand-built +28% vs systematic -13%), Hyundai Motor and Toyota, where the hand-built models normalise depressed ROE. (c) **Did the screen throw away winners?** Running the systematic model on all 288 in-scope names finds only 4 that clear the hurdle but were not shortlisted (SK Innovation, HMM, Sony, Toyota Tsusho), all by 20–30%. The screen costs little.

3 · From 50 candidates to 19 holdings

A single max-Sharpe run is fragile: small changes in expected returns swap names in and out. So the optimiser is run 60 times, each time adding random noise to every expected return (± 3 pp a year for stocks, ± 1 pp for ETFs), under the full mandate (hedges $\geq 38\%$, sector caps, market cap $\leq 60\%$). A stock is held only if it gets at least 0.5% in half the runs or more (Michaud-style resampling). The chart shows the outcome.



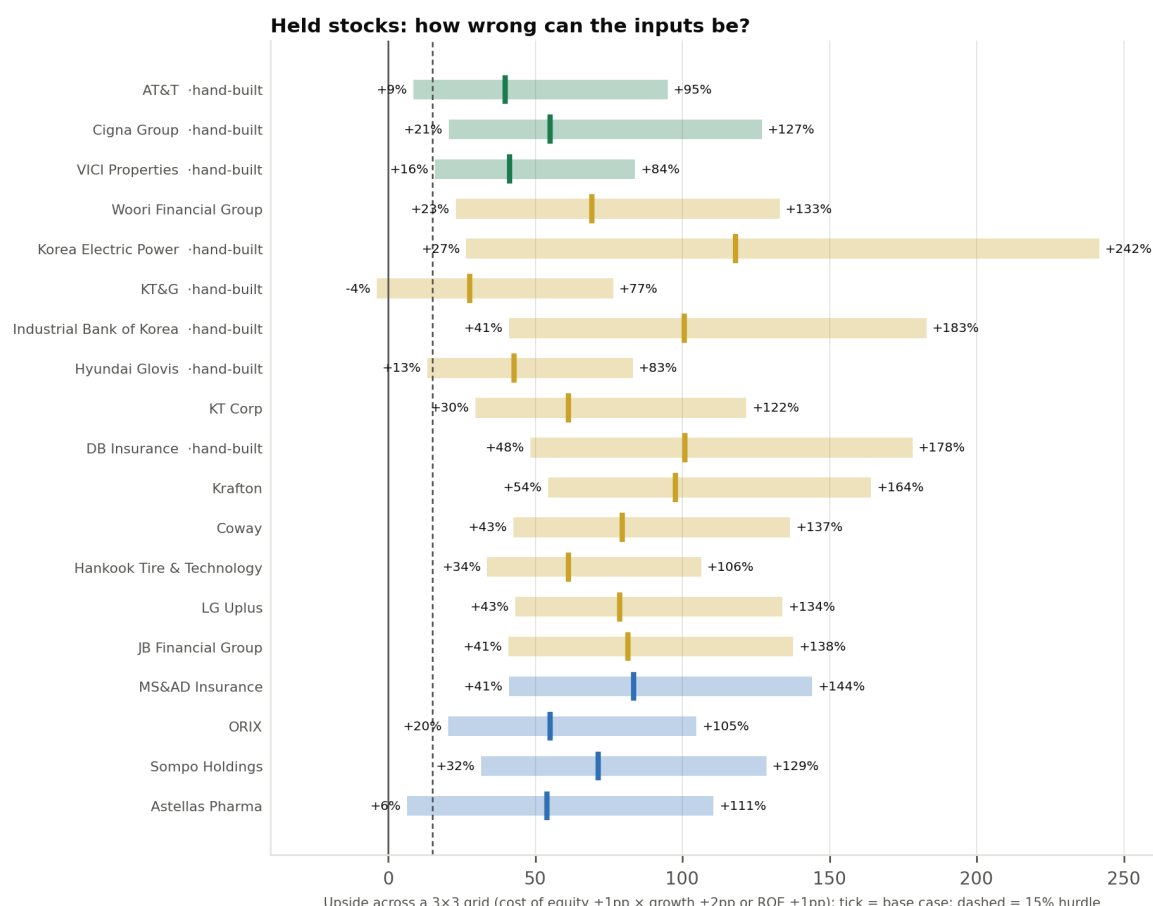
Why familiar names dropped out. Upside is not the only thing that matters: the optimiser also looks at risk and overlap. **US stocks** carry USD/KRW risk for a Korean investor and have a lower assumed equity premium, so a US name needs more upside to earn a place; UnitedHealth (+52%) also competes with Cigna under a 6% managed-care cap. **KB, Shinhan and Hana** pass the hurdle but the 9% cap on Korean financials is filled by IBK, Woori, JB Financial and DB Insurance, which are cheaper. **Samsung Pref.** (+15%) and **Toyota** (+24%) clear the hurdle only narrowly, and Toyota's systematic value is 0%. **PayPal** (+40%) is high-beta. None of these is a sell call; they are simply not the best use of a limited risk budget.

4 · The 19 stocks held

Holding	Code	Wt	Model	Key inputs	Price	Value	Upside
United States · 3 stocks · 5.0%							
AT&T †	T	2%	FCFE	2026 FCF guidance \$18bn+ (10% yield); 0% growth; Ke 7.3%	\$25.38	\$35.49	+40%
Cigna Group †	CI	1.75%	FCFE	FCF \$8.0bn (11% yield); +4%/yr 5 yrs then 0% forever; Ke 8.4%	\$274.47	\$425.44	+55%
VICI Properties †	VICI	1.25%	AFFO model	AFFO \$2.51 (P/AFFO 9.4x); g 2%→1.5%; k 9.3%	\$23.71	\$33.51	+41%
Korea · 12 stocks · 28.5%							
Woori Financial Group	316140	2%	Justified P/B	ROE 8.8% (trailing/fwd avg), k 8.8%, g 2.5%	₩33,200	₩56,199	+69%
Korea Electric Power †	015760	2.75%	Justified P/B	Normalised ROE 7% (trailing 15.9%); k 7.7%; g 1.0%	₩30,250	₩65,974	+118%
KT&G †	033780	2.75%	Justified P/B	ROE 14.1%; k 7.2%; g 2.5%	₩174,300	₩222,502	+28%
Industrial Bank of Korea †	024110	2.25%	Justified P/B	ROE 7.4% (P/B+P/E); k 8.3%; g 2.0%	₩20,400	₩40,932	+101%
Hyundai Glovis †	086280	1.5%	Justified P/B	ROE 15.3% (P/B+P/E); k 9.4%; g 3.0%	₩196,900	₩281,023	+43%
KT Corp	030200	2.5%	Earnings/FCF model	earnings/FCF blend; g 0.7% (½ consensus 3y -1% + ½ terminal) → 0.7%; ROE 8%; k 8.8%	₩53,400	₩86,168	+61%
DB Insurance †	005830	2.75%	Justified P/B	ROE 10.9% (P/B+P/E); k 7.7%; g 2.5%	₩184,600	₩370,687	+101%
Krafton	259960	2.5%	Earnings/FCF model	earnings/FCF blend; g 7.0% (½ consensus 3y 19% + ½ terminal) → 2.5%; ROE 12%; k 8.8%	₩200,000	₩395,322	+98%
Coway	021240	2.5%	Earnings model	earnings only (FCF ≤ 0); g 7.2% (½ consensus 3y 12% + ½ terminal) → 2.5%; ROE 19%; k 8.8%	₩95,200	₩170,910	+80%
Hankook Tire & Technology	161390	2.5%	Earnings/FCF model	earnings/FCF blend; g 5.0% (½ consensus 3y 15% + ½ terminal) → 2.5%; ROE 8%; k 8.8%	₩63,500	₩102,461	+61%
LG Uplus	032640	2.5%	Earnings/FCF model	earnings/FCF blend; g 4.8% (½ consensus 3y 20% + ½ terminal) → 2.5%; ROE 8%; k 8.8%	₩15,020	₩26,836	+79%
JB Financial Group	175330	2%	Justified P/B	ROE 12.4% (trailing/fwd avg), k 8.8%, g 2.5%	₩29,050	₩52,756	+82%
Japan · 4 stocks · 8.8%							
MS&AD Insurance	8725	2.25%	Justified P/B	ROE 12.3% (trailing/fwd avg), k 7.1%, g 1.5%	¥5,030	¥9,223	+83%
ORIX	8591	2%	Justified P/B	ROE 13.4% (trailing/fwd avg), k 7.1%, g 1.5%	¥6,152	¥9,536	+55%
Sompo Holdings	8630	2.25%	Justified P/B	ROE 12.1% (trailing/fwd avg), k 7.1%, g 1.5%	¥6,888	¥11,800	+71%
Astellas Pharma	4503	2.25%	Earnings/FCF model	earnings/FCF blend; g 1.9% (½ trailing→fwd EPS 2% + ½ terminal) → 1.5%; ROE 21%; k 7.1%	¥2,399	¥3,696	+54%

† hand-built model from the earlier version; the rest use the systematic model on page 3. Price and value in local currency.

5 · How wrong can the inputs be, and what the screen found



United States · 3 stocks, 5.0% of the portfolio

Pattern	Only 13 of 39 valued US names clear the hurdle; the US market is the most fully priced of the three. US equity exposure comes mainly through IVV (5%), with single names only where the discount is large.
Cigna, AT&T, VICI	Cigna: 8.6x forward earnings, valued with zero growth after year 5 to price PBM reform. AT&T: \$18bn+ FCF guidance, a ~10% yield, valued at zero growth. VICI: 9.4x AFFO with organic growth only.
Rejected	Microsoft and Alphabet (both about -25 to -55% on either model), Apple, NVIDIA, JPMorgan and the other mega-caps: priced for a lower discount rate than 5.23% implies. They are owned at index weight inside IVV.

Korea · 12 stocks, 28.5% of the portfolio

Pattern	The Korean survivors are not the index heavyweights. They are cash-generative domestic businesses trading below book or at single-digit P/E: telcos, lenders, insurers, a utility, and three consumer names. This is the 'Korea discount' in its purest form, and the Value-up programme is the catalyst.
Telcos	KT and LG Uplus trade at 0.6–0.7x book with 4.5% dividends and very high FCF yields. Consensus expects EPS growth as depreciation from 5G capex rolls off; the model uses only half of it. Telecom is capped at 8% across all markets (now 7.0%).
Financials	IBK, Woori, JB Financial and DB Insurance fill the 9% cap on Korean financials (9.0%). IBK and DB Insurance show +100%: IBK's cheapness partly reflects state ownership, and IFRS 17 makes insurer book values less comparable, so treat those as upper bounds.
KEPCO	The largest model upside (+118%) and the widest error band. The model uses a normalised 7% ROE, well below the trailing 16%. Tariffs are set politically, so it is capped at 3%, and it loses in an oil spike.
Krafton, Coway, Hankook	Krafton: 8.5x forward earnings with net cash; the risk is dependence on one franchise (PUBG). Coway: 19% ROE at 9x earnings; FCF is negative because rental units are capitalised, so it is valued on earnings only. Hankook: 5.4x forward P/E at half of book, with tariff risk on US sales.
KT&G, Glovis	Carried over from the earlier version on their hand-built models. Note that KT&G's systematic value is below the price; it is held on the hand-built model and a low beta.

Japan · 4 stocks, 8.8% of the portfolio

Pattern	Two non-life insurers (MS&AD, Sampo), ORIX and Astellas. The insurers benefit directly from higher JGB yields on their float and are unwinding cross-shareholdings, which lifts ROE. The megabanks (MUFG, SMFG) have already re-rated.
Astellas	11.6x forward earnings and a 21% ROE; the risk is the patent cliff on its prostate-cancer drug, which the model handles only through low growth (1.9%).
How to buy	Tokyo trades in 100-share lots (≈¥2–6m each at ¥8.6¥). Hitting these weights exactly needs about ¥2.6¥ in total. ORIX also trades on the NYSE (ADR 'IX'); with a smaller account, hold the Japan stock sleeve through the TOPIX ETF instead.
Dropped	Toyota (held in the earlier version via the TM ADR) passes the hurdle only on its hand-built model (+24%) and is not selected. Nippon Steel and NTT came closest (38% of runs).

6 · ETFs: core beta, real assets and a 42.0% hedge sleeve

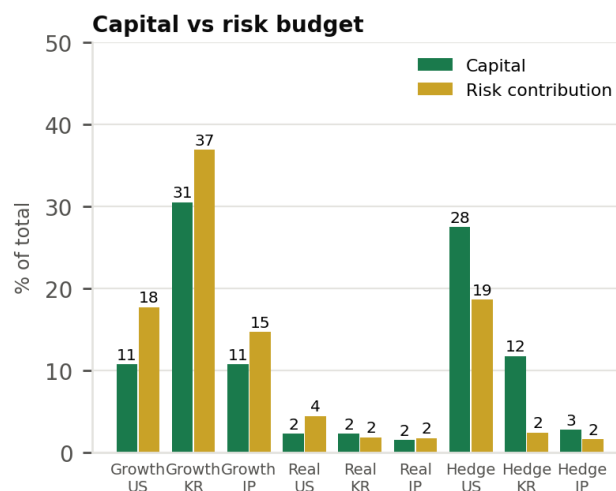
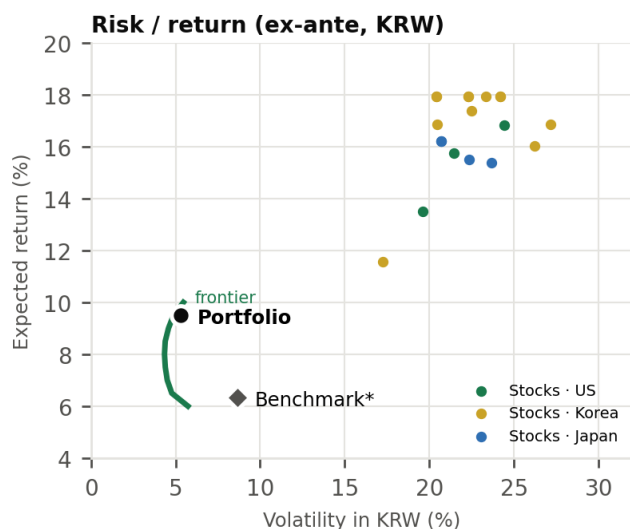
The hedge design is **one instrument per type of crisis**, because 2022 proved that bonds alone fail when inflation is the shock. For a KRW-based investor there are two further hedges: the dollar and the yen. In every Korean crisis on record (1997, 2008, 2020) the won fell as stocks fell, so the 56% of the book held in USD and JPY assets rises in KRW exactly when Korean assets drop. The ETF list is unchanged from the earlier version; only the weights moved.

ETF	Wt	Exposure	Protects against / role	Carry	When it fails
IVV 069500 · KR	5%	Core US equity	—	5.0–6.5%	US bear market
USMV	1%	Lower-beta US equity	Sharp sell-offs (falls ~70% as much)	≈6%	Momentum / tech rallies
XLU	1%	US utilities	Defensive; rate-sensitive income	≈6.5%	Sharp rate rises
KODEX 200 069500 · KR	2%	Core Korea equity	—	≈8.5%	Korea-specific shock
TOPIX ETF 1306 · JP	2%	Core Japan equity	—	≈7.5%	Strong yen hurts exporters
VNQ	1%	Broad US REITs	Inflation via rents	≈7%	Rate spikes
J-REIT ETF 1343 · JP	1.5%	Japanese REITs	Rent reflation	≈4.5% distr. (est.)	JGB yield spikes
TIGER REITs 329200 · KR	2.25%	Korean REITs & infrastructure	Inflation via rents / tolls	≈9.7% distr.	KRW rate spikes
IEF	2.5%	7–10y US Treasuries	Recession, credit crash (+18% in 2008)	≈5.0% YTM	Inflation shock (2022: ~15%)
KODEX KTB10 152380 · KR	5.75%	10y Korean govt bond futures	Korean recession / BOK cuts	≈4.4%	KRW rate rises
KODEX KTB3 114260 · KR	1.5%	3y Korean govt bonds	Low-duration KRW ballast	≈3.3%	BOK hikes (mildly)
JGB ETF 2561 · JP	2.75%	Japanese govt bonds	Yen safe-haven + duration	≈2.5% + yen	Yen weakness
VTIP	2.75%	0–5y US TIPS	Unexpected inflation, little duration	Real yld + CPI	Disinflation
PDBC	1.5%	Diversified commodities	Oil / Middle-East spike (+~20% in 2022)	Collateral + roll	Global recession
IAU	1.5%	Gold	Geopolitics, Fed credibility, USD debasement	None	Rising real yields
KODEX Gold(H) 132030 · KR	1.5%	Gold futures, FX-hedged to KRW	Gold without adding more USD	–1.4% hedge cost	Rising real yields
DBMF	3.75%	Managed futures (trend)	Stocks & bonds falling together (CTA index ≈+20% in 2022)	T-bill + trend	Choppy, trendless markets
KMLM	3.25%	Managed futures (different trend model)	Diversifies DBMF's model risk	T-bill + trend	Choppy, trendless markets
TAIL	4.5%	Put options + T-notes	Crash convexity (sudden ~20% equity)	Negative (option bleed)	Slow grind lower; rallies
BTAL	4.5%	Long low-beta / short high-beta	Speculative-growth unwind; $\beta \approx -0.5$	≈Cash	Junk / high-beta rallies
KODEX Inverse 114800 · KR	1%	Short KOSPI 200 (daily inverse)	Korea-specific crash	≈ –3% (short equity)	Any Korean rally; daily reset
KODEX USD 261240 · KR	1%	Long USD vs KRW futures	KRW crisis	≈ –1.2% (rate gap)	KRW appreciation
SGOV	3.25%	0–3m US T-bills	Rate hikes; funds rebalancing	≈4.2%	Rapid Fed cuts
KODEX CD 459580 · KR	1%	KRW CD-rate fund	KRW liquidity	≈2.8% (est.)	BOK cuts

Why hedges get 42.0% of capital but only 23% of the risk. Most hedges are low-volatility and load negatively on equity, or positively on factors that rise in a crisis: duration, gold, USD and JPY vs KRW. The diversification ratio is **3.19** (weighted-average asset volatility ÷ portfolio volatility). The price is paid in bull markets: TAIL bleeds option premium, BTAL loses in speculative rallies, and a strong won hurts every USD and JPY asset (page 9).

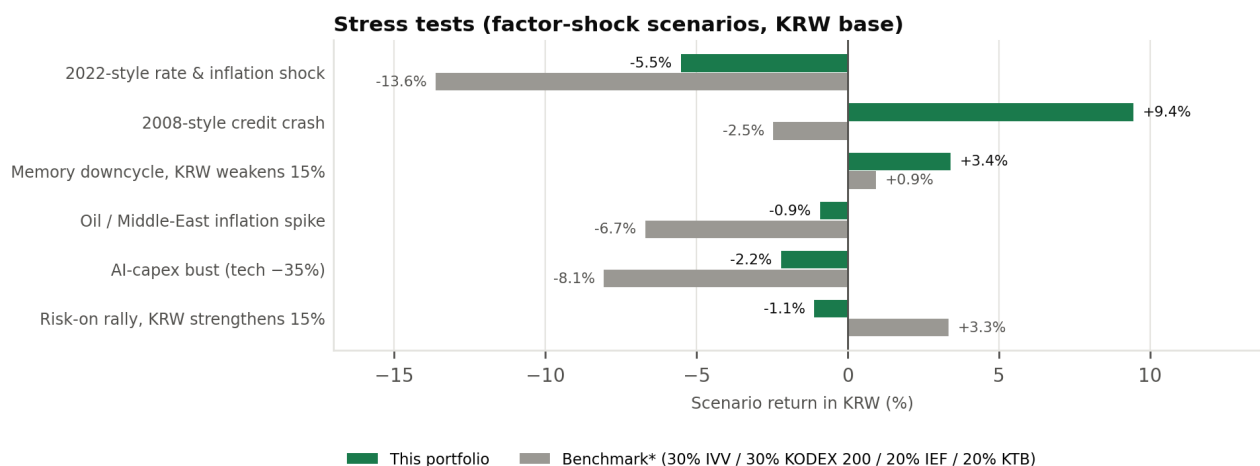
7 · Portfolio construction and risk

Two passes. Pass 1 (selection) is the resampled max-Sharpe on page 4. Pass 2 (sizing) maximises Sharpe over KRW cash with a risk-parity penalty on the selected names: $\min -SR(w) + \lambda \cdot N \cdot \Sigma(RC_i - \text{mean } RC)^2$, $\lambda = 0.5$. **Constraints:** long-only; single stocks 0.8–3%; REITs 0.8–3%; IVV 5–12%, KODEX 200 and TOPIX 2–6%; other funds 1–7%; growth sleeve 45–52%, real assets 6–12%, hedges ≥38%; **no market floors**, each market ≤60%; KR financials ≤9%; JP financials ≤9%; US financials ≤6%; Telecoms (all markets) ≤8%; Autos (all markets) ≤8%; US managed care / pharmacy ≤6%; Energy & shipping ≤5%; Trading houses & steel ≤5%. Covariance comes from an 11-factor model (page 11). SLSQP, rounded to 0.25%.



Ex-ante (KRW)	Portfolio	Benchmark*
Expected return	9.5%	6.3%
Volatility	5.3%	8.7%
Sharpe (vs 2.8% KRW cash)	1.26	0.41
95% VaR / CVaR (1 yr)	0.8% gain / 1.5%	7.9% / 11.6%
Market split US / KR / JP	41 / 45 / 15%	50 / 50 / 0%
Holdings / largest weight	43 / 5.8%	4 / 30%

Reading the results. Normal-distribution VaR comes out as a small gain because the model expects 9.5% against 5.3% volatility. Treat that as a sign the expected returns are optimistic, not as a guarantee: the stress tests below are the better guide. In a 2008-style crash the portfolio *gains* in KRW because the won collapses while dollar and yen assets rally; that depends on currency relationships holding. **The cost:** a risk-on rally with a 15% stronger won returns -1.1% versus +3.3% for the benchmark.

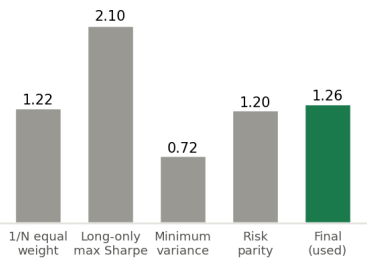


8 · Does the optimiser add value? Five weighting methods compared

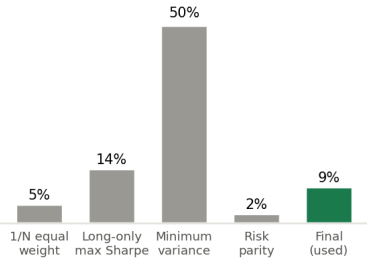
Same 43 assets, same risk model, same expected returns, five ways to set the weights. This is the test any optimiser has to pass: it must beat naive diversification by enough to justify the estimation risk it takes on.

Method	E[R]	Vol	Sharpe	Max wt	Max risk	Eff. N	Hedges	Verdict
1/N equal weight	9.7%	5.7%	1.22	2.3%	5%	43	37%	Zero estimation error; breaks the ≥38% hedge floor
Long-only max Sharpe	10.7%	3.8%	2.10	28.1%	14%	9	31%	Best on paper; ~9 effective bets, below the hedge floor
Minimum variance	3.9%	1.6%	0.72	49.8%	50%	3	74%	Half the book in one hedge; return too low
Risk parity	8.3%	4.6%	1.20	14.3%	2%	23	52%	Competitive without using any valuation views
Final (mandate + RP penalty)	9.5%	5.3%	1.26	5.8%	9%	35	42%	Used: meets every mandate limit

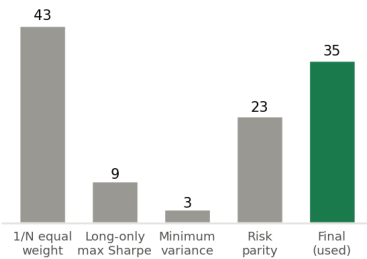
Sharpe ratio (ex-ante)



Largest single risk contribution



Effective no. of holdings (1/Σw²)



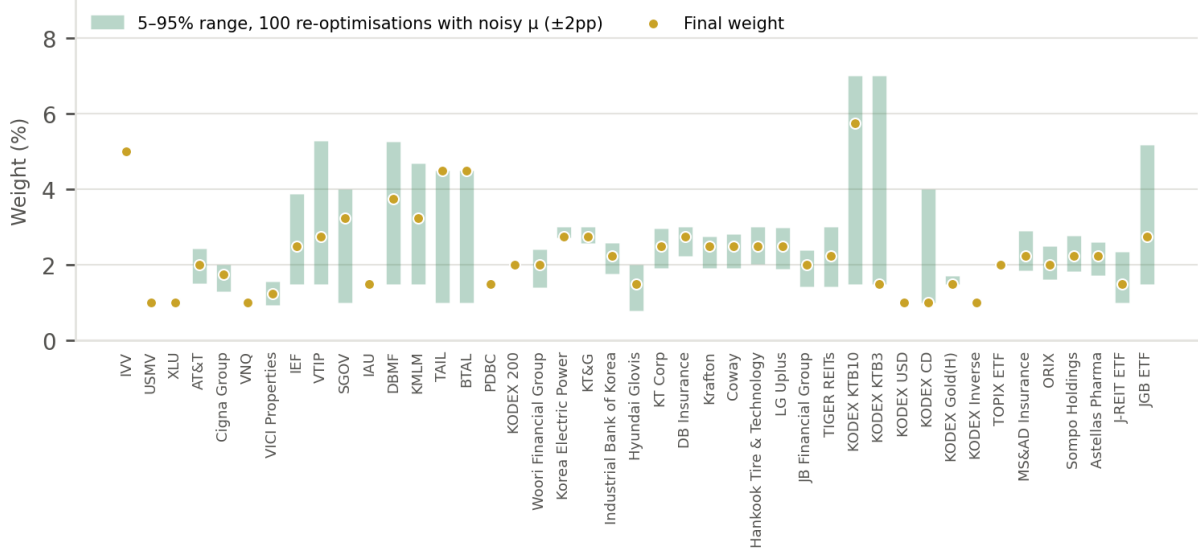
Result. Equal-weighting the same 43 assets scores a Sharpe of 1.22, against 1.26 for the final portfolio, and risk parity reaches 1.20 without using a single valuation. So, as in the earlier version, **most of the value comes from choosing the assets, not from the sizing step** (DeMiguel, Garlappi & Uppal, 2009). The difference now is that the choice itself is systematic: screen, value, resample. The sizing step adds mandate compliance: hedges at 42% (1/N only reaches 37%) and no position above 6%. The unconstrained Markowitz solution promises a Sharpe of 8.2 but needs 20x gross leverage: an artefact of estimation, not an opportunity.

Robustness checks

λ (risk-parity penalty)	Sharpe	Vol	Max risk
0	1.38	4.8%	10%
0.25	1.29	5.3%	9%
0.5 (used)	1.26	5.3%	9%
1	1.25	5.3%	9%
5	1.17	5.3%	9%

Penalty strength. Moving λ from 0 (pure max-Sharpe inside the mandate) to 5 (near risk parity) changes the Sharpe only from 1.38 to 1.17. **Noisy expected returns.** Re-optimising 100 times with ±2pp errors in every μ moves weights by 1.4pp on average (5–95% band). Single-stock weights barely move because their caps bind; the real uncertainty is **the hedge mix** (bond, TIPS, trend and tail-hedge funds), so those weights should be read as ranges.

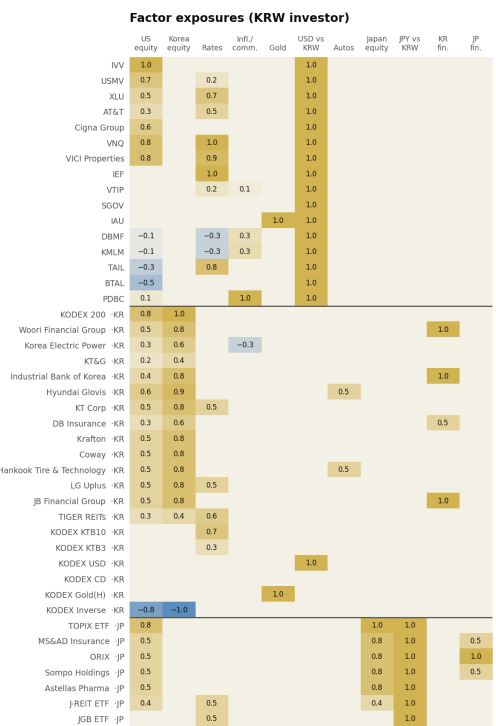
How much do weights move when expected returns are wrong?



9 · Methodology, assumptions and sources

Factor	Vol	Key correlations (assumed)	What loads on it
US equity	16%	−0.35 with USD/KRW; +0.20 with inflation	US stocks (β), IVV, USMV, REITs; 0.6×β for Korean and Japanese stocks; negative for TAIL, BTAL
Korea equity (Korea-specific)	14%	−0.45 with USD/KRW	Korean stocks (β), KODEX 200, K-REITs
Rates (duration return)	7%	−0.30 with inflation; +0.25 with gold	IEF, KTB, JGB, TAIL, REITs, telcos and utilities (0.5)
Inflation / commodities	18%	+0.35 with gold	PDBC, DBMF, KMLM, VTIP; negative for KEPCO
Gold	15%	+0.10 with USD/KRW	IAU, KODEX Gold(H)
USD vs KRW	9%	Rises when equities fall	Every US-listed asset (unhedged) + KODEX USD futures
Autos	12%	Independent	Hyundai Glovis (0.5)
Japan equity (Japan-specific)	13%	−0.30 with JPY/KRW	TOPIX ETF, Japanese stocks, J-REITs
JPY vs KRW	11%	+0.55 with USD/KRW; −0.40 with Korea equity	Every Japanese asset (unhedged)
Korean financials	12%	+0.30 with Japanese financials	IBK, Woori, JB Financial; DB Insurance at 0.5
Japanese financials	13%	−0.30 with rates (banks gain as yields rise)	ORIX; MS&AD and Somo at 0.5

Covariance. $\Sigma = B \cdot F \cdot B^T + D$, where B holds each asset's factor loadings, F is the factor covariance and D holds stock-specific variance (13–25% vol for stocks, higher for betas above 1.3 and for cyclicals). Loadings for the new names are set by rule from their data-feed beta, market and sector; the earlier version's hand-set loadings are kept where they exist. **Expected returns.** Stocks: CAPM cost of equity + half the valuation gap closed over 3 years, with the upside counted at most at 60% so that no single model can dominate. Funds: yield-based or long-run premia (IVV 6.5%, KODEX 200 8.5%, IEF 5.0%, VTIP 4.7%, SGOV 4.2%, gold 4.0%, DBMF/KMLM 5.5%, PDBC 4.5%, TAIL 1.0%, BTAL 2.0%, TOPIX 7.5%, JGB 2.5%, KODEX USD −1.2%, KODEX Inverse −3%). FX is assumed to drift 0. **Stress tests** apply factor shocks anchored to 2008 and 2022, with overrides where an asset behaves non-linearly (TAIL, DBMF, BTAL, KEPCO in an oil spike).



Limitations. The screen uses one data vendor's multiples at one date; a bad data point can move a name in or out. The systematic model is deliberately simple: in the payout formula ROE is floored at 8%, which flatters low-ROE telcos; consensus growth is halved but still drives the high-upside names; and Korean upsides of 60–120% mainly restate the long-standing 'Korea discount', which may simply persist. Volatilities, correlations and loadings are assumptions anchored to history, not estimates from a fresh price download. Normal VaR understates fat tails. KOSDAQ stocks and Japanese mid-caps are outside the universe. The market split (Korea-heavy) follows from cheaper Korean valuations, a higher assumed Korean equity premium and the absence of FX risk for a KRW investor; a reader who disagrees with any of those should cap Korea.

Sources

Screen data: Stock Analysis (stockanalysis.com) statistics pages for all 300 stocks (price, market cap, trailing and forward P/E, P/B, P/FCF, ROE, dividend yield, beta, free cash flow, net cash, 3-yr EPS growth consensus), 22–28 Sep 2026. Hand-built models: sources as listed in the earlier version (Tickzen, Stock Titan, Judal, Chickstock, Daum/FnGuide, company guidance). Macro: Washington Trust weekly review (25 Sep 2026), Trading Economics (KTB and JGB 10-yr), Capital.com (BOJ). Method: DeMiguel, Garlappi & Uppal (2009), 'Optimal versus naive diversification', *Review of Financial Studies*; Michaud (1998), *Efficient Asset Management*.